up onto the volcano's rim, but Mann's planet does not quite make it up to the rim; it spirals back down the volcano's side (centrifugal forces push it outward) and then up the gravitational energy surface, away from Gargantua.

50 The agreement between our volcano-rim analogy and these force arguments is due to a key fact: The net force (gravitational plus centrifugal) on the *Endurance* is proportional to the slope of the energy surface (Figures 27.3 and 27.5). Can you figure out why?